

HAT YAI INTL, Thailand

WMO: 485690

Lat: 6.933N

Lon: 100.393E

Elev: 27

StdP: 101.00

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	21.4	22.2	18.7	13.6	28.9	19.4	14.2	28.4	8.1	30.1	7.4	29.9	0.3	270	0.317

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	9.7	35.1	25.5	34.4	25.6	33.9	25.6	27.6	31.6	27.2	31.4	26.8	31.1	3.9	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.7	22.4	30.1	26.1	21.6	29.3	25.8	21.2	29.0	88.3	31.7	86.0	31.7	84.4	31.3	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	6.3	5.5	19.9	36.6	1.3	1.0	18.9	37.4	18.2	38.0	17.4	38.6	16.5	39.3		
(5)			19.1	28.6	1.3	0.8	18.1	29.2	17.4	29.7	16.6	30.1	15.7	30.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.9	26.7	27.4	28.4	29.0	28.9	28.6	28.3	28.2	27.9	27.6	27.0	26.5	(6)
(7)		DBStd	1.25	0.96	0.96	0.93	0.96	0.91	0.89	0.87	0.96	0.89	0.88	1.13	1.11	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6532	519	487	570	571	587	558	568	565	538	546	511	512	(10)
(11)		CDD18.3	3490	261	254	312	321	329	308	309	307	288	288	261	254	(11)
(12)		CDH23.3	33921	2311	2517	3262	3428	3493	3224	3186	3070	2698	2506	2183	2043	(12)
(13)		CDH26.7	13288	759	1009	1466	1566	1492	1348	1316	1261	1007	851	648	565	(13)
(14)	Wind	WSAvg	1.6	2.0	2.2	1.8	1.3	1.4	1.5	1.6	1.7	1.5	1.3	1.3	1.7	(14)
(15)	Precipitation	PrecAvg	1794	76	39	100	127	130	124	117	140	158	234	289	245	(15)
(16)		PrecMax	2603	366	160	347	282	305	243	211	313	251	433	696	658	(16)
(17)		PrecMin	1269	8	1	4	11	57	15	38	34	83	116	116	39	(17)
(18)		PrecStd	284	84	43	80	66	49	55	45	58	47	74	139	139	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.4	34.2	35.8	36.6	36.0	35.0	34.8	35.0	34.1	33.7	33.0	32.2	(19)
(20)			MCWB	25.1	24.4	24.7	25.3	25.8	25.7	25.5	25.3	25.6	25.7	25.9	25.6	(20)
(21)		2%	DB	31.8	33.1	34.6	35.2	34.8	34.0	33.9	33.9	33.2	32.8	32.0	31.6	(21)
(22)			MCWB	24.9	24.4	25.0	25.6	26.0	25.8	25.5	25.4	25.6	25.8	25.8	25.4	(22)
(23)		5%	DB	31.1	32.2	33.8	34.2	33.9	33.2	33.1	33.1	32.6	32.0	31.1	30.9	(23)
(24)			MCWB	24.6	24.4	25.0	25.7	26.2	25.9	25.5	25.4	25.6	25.8	25.8	25.1	(24)
(25)		10%	DB	30.4	31.7	32.9	33.2	32.9	32.5	32.2	32.2	31.8	31.0	30.2	29.9	(25)
(26)			MCWB	24.5	24.4	25.0	25.9	26.2	25.9	25.5	25.5	25.6	25.8	25.7	25.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.0	26.5	27.6	28.0	28.2	27.7	27.6	27.7	27.6	27.6	27.6	27.1	(27)
(28)			MCD B	30.3	31.0	32.1	32.4	32.0	31.9	31.6	31.6	31.4	31.1	30.5	30.1	(28)
(29)		2%	WB	26.1	25.7	26.5	27.1	27.3	27.1	26.7	26.7	26.7	26.7	26.8	26.2	(29)
(30)			MCD B	29.7	30.4	31.7	32.1	31.8	31.5	31.0	31.1	30.7	30.5	29.8	29.4	(30)
(31)		5%	WB	25.5	25.3	26.0	26.6	26.9	26.6	26.3	26.2	26.2	26.3	26.3	25.9	(31)
(32)			MCD B	29.2	29.9	31.3	31.6	31.5	31.1	30.8	30.7	30.4	30.0	29.4	29.0	(32)
(33)		10%	WB	25.1	24.9	25.6	26.2	26.5	26.2	26.0	25.9	25.9	26.0	25.9	25.4	(33)
(34)			MCD B	28.8	29.5	30.6	31.1	31.0	30.7	30.5	30.4	30.0	29.7	29.1	28.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.8	9.3	10.0	9.7	8.9	8.6	8.6	8.1	7.6	6.5	6.4	(35)	
(36)			MCDBR	8.6	10.1	11.2	10.8	9.9	9.5	9.6	9.7	9.3	8.7	8.1	8.1	(36)
(37)			MCWBR	3.2	3.2	3.4	3.2	3.3	3.3	3.3	3.4	3.3	3.4	3.3	3.1	(37)
(38)		5% WB	MCDBR	7.6	9.2	9.8	9.8	9.0	8.8	8.7	8.8	8.4	7.9	7.1	6.8	(38)
(39)			MCWBR	3.5	3.4	3.5	3.6	3.5	3.5	3.4	3.6	3.5	3.5	3.4	3.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.459	0.466	0.475	0.466	0.451	0.459	0.464	0.462	0.453	0.451	0.454	0.462	(40)	
(41)		taud	2.289	2.247	2.246	2.279	2.346	2.350	2.326	2.316	2.340	2.349	2.339	2.301	(41)	
(42)		Ebn at Noon	858	863	855	849	843	826	825	841	861	865	857	848	(42)	
(43)		Edh at Noon	135	144	145	138	126	124	127	132	130	128	127	131	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.86	5.80	5.87	5.77	5.17	4.96	4.88	5.00	4.87	4.61	4.20	4.11	(44)	
(45)		RadStd	0.36	0.34	0.47	0.27	0.27	0.22	0.22	0.17	0.18	0.32	0.34	0.43	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+75	+76
(47) Regional (16 neighbors)		+0.22	+0.15	N/A	+0.32	+0.21	+0.19	N/A	N/A	+94	+94

Nomenclature: See separate page